

MANIS CHAUDHARY

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SUMMARY

Data Science and Artificial Intelligence professional with expertise in Machine Learning, Deep Learning, Computer Vision, NLP, Predictive Analytics, and Data Engineering. Experienced in developing AI-driven solutions, machine learning workflows, computer vision systems, and predictive models using Python, SQL, TensorFlow, PyTorch, OpenCV, and AWS through software engineering experience and advanced technical projects. Strong interests in AI for Cybersecurity, Reasoning LLMs, and Intelligent Systems.

EXPERIENCE

Software Engineer

Summus Engineering and Consultancy Pvt. Ltd, Kathmandu, Nepal

July 2022 – June 2024

- Developed and maintained software solutions supporting data processing, SQL databases, and business intelligence workflows, improving operational efficiency and system reliability.
- Optimized SQL queries, relational database structures, and backend workflows, enhancing system performance and enabling efficient data retrieval for business operations.

Teaching Assistant

The British College, Kathmandu, Nepal

September 2022 – September 2023

- Assisted in research, analytical problem-solving, and project-based academic activities, supporting coursework and structured technical learning.
- Provided technical mentoring and academic guidance to students, strengthening communication, analytical thinking, and problem-solving skills.

IT Officer Intern

Devsuits Pvt. Ltd, Kathmandu, Nepal

November 2021 – December 2022

- Conducted data collection, online research, and analytical support for internationally funded projects across Nepal, Liberia, and South Sudan.
- Supported web-based information systems, technical documentation, and digital coordination for projects funded by EU, GIZ, UN, and INGOs.

TECHNICAL SKILLS

Programming: Python, SQL, R, Java, JavaScript, TypeScript, PHP, C++, HTML/CSS

Machine Learning & Artificial Intelligence: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), Predictive Modeling, Classification, Regression, Model Training, Model Evaluation, Feature Engineering, Artificial Intelligence, Reasoning LLMs

Frameworks & Libraries: TensorFlow, PyTorch, Scikit-learn, OpenCV, Pandas, NumPy, Matplotlib, Keras, Spark/PySpark, Hadoop, YOLOv8

Data Science & Analytics: Exploratory Data Analysis (EDA), Data Cleaning, Data Preprocessing, Statistical Analysis, Predictive Analytics, Data Visualization, ETL, Data Engineering

Cloud & Databases: AWS, Google Cloud Platform (GCP), MySQL, MongoDB, SQL Optimization, Database Management, Airflow

Tools & Technologies: Git, GitHub, Docker, CI/CD, Jupyter Notebook, Google Colab, VS Code, REST APIs, Agile/Scrum, Gradio, Streamlit

EDUCATION

University of New Haven

Master of Science in Data Science

West haven, CT

(January 2025 – Dec 2026)

Leeds Beckett University

Bachelor of Science in Computing (Hons)

Leeds, United Kingdom

(Sept 2018 – July 2022)

PROJECTS

Real-Time Monocular Visual SLAM using ORB-SLAM3 & MonoGS — Computer Vision Project

- Developed a real-time monocular visual simultaneous localization and mapping (SLAM) system using ORB-SLAM3, MonoGS, Python, and OpenCV for trajectory estimation, localization, and environment mapping.
- Evaluated algorithm performance using benchmark datasets and custom closed-loop environments, applying feature extraction, motion estimation, and localization techniques for real-time mapping accuracy.

NYC Subway Sign & Digital Display Detection — YOLOv8 Computer Vision Project

- Built a YOLOv8-based object detection system using Python, OpenCV, and PyTorch to detect NYC subway signs and digital display boards under varying subway conditions.
- Generated a synthetic dataset using generative AI techniques, including annotation, preprocessing, model training, evaluation, and developed a Gradio-based interface for real-time inference and visualization.

Healthcare Fraud Detection System — Machine Learning Project

- Developed a healthcare fraud detection system using AWS, PySpark, Python, and Random Forest for fraud classification and predictive analytics.
- Applied data preprocessing, feature engineering, model training, and evaluation, achieving 94% classification accuracy.Optimizing Taxi Services with Predictive Analysis — Predictive Analytics Project

Optimizing Taxi Services with Predictive Analysis — Predictive Analytics Project

- Developed a predictive analytics model for taxi trip duration estimation using real-world transportation datasets and machine learning techniques.
- Performed EDA, feature engineering, preprocessing, and model evaluation to improve prediction performance.